

31. $\frac{12}{\sqrt{40}}$
 32. $\frac{20}{\sqrt{150}}$
 33. Determine the conjugate of $5 + \sqrt{6}$.
 34. Determine the conjugate of $10 - \sqrt{13}$.
 35. Determine the conjugate of $4 + 3\sqrt{5}$.
 36. Determine the conjugate of $5\sqrt{15} + 8\sqrt{13}$.
 37. Find the product of $3 + 2\sqrt{7}$ and its conjugate.
 38. Find the product of $2\sqrt{3} - 5$ and its conjugate.

For the following exercises, rationalize the denominator of the fraction, and then simplify the fraction.

39. $\frac{4}{1 + \sqrt{3}}$
 40. $\frac{6}{5 + \sqrt{7}}$
 41. $\frac{-4}{5 - \sqrt{10}}$
 42. $\frac{10}{\sqrt{6} + 7}$

3.6 Real Numbers

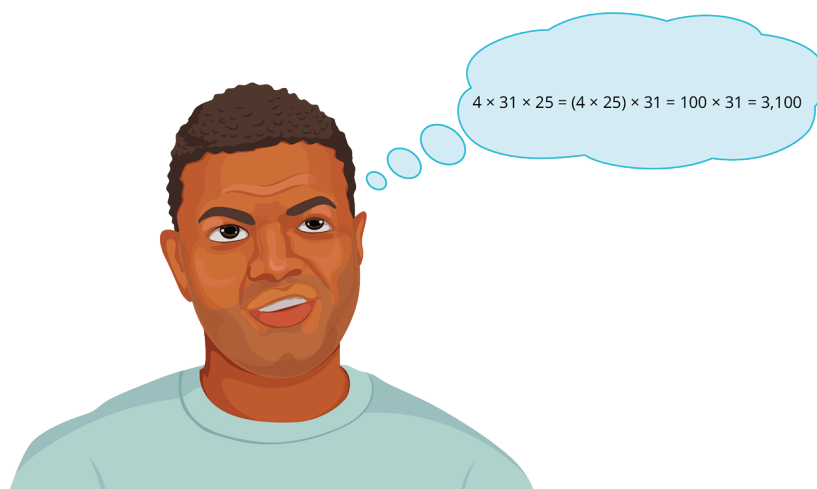


Figure 3.31 Quick mental math involves using the known properties of real numbers.

Learning Objectives

After completing this section, you should be able to:

1. Define and identify numbers that are real numbers.
2. Identify subsets of the real numbers.
3. Recognize properties of real numbers.

Have you ever been impressed by the speed at which someone can do math in their head? Most of us at one time or another have witnessed a person speed through mental math, an impressive feat that often bests calculators. One such person is Neelkantha Bhanu Prakash. As of September 20, 2020, he is considered the world's fastest human calculator. He currently holds four world records. How does someone do that, though? Have they memorized lots of arithmetic facts? Are they simply brilliant?

The answer isn't simple so much as it is about knowledge. Real numbers behave in some very regular ways, following rules that can be learned. In this section, those rules are explored.

Watch the video of Arthur Benjamin's TED Talk to learn about another mathematician with remarkable mental abilities.

VIDEO

Arthur Benjamin TED Talk, *Faster Than a Calculator* (https://openstax.org/r/Arthur_Benjamin_TED_talk,_Faster_than_a_Calculator)

Defining and Identifying Real Numbers

Real numbers are the rational and irrational numbers combined. The real numbers represent the collection of all physical distances that exist, along with 0 and the negatives of those physical distances. For example, if you take a measure of three units, and divide that distance into eight (8) equal lengths, the distance you have formed is $\frac{3}{8}$ units.

Also, if you draw a **right triangle** (a triangle with one angle equal to 90 degrees) with one side length of 1, and the other side length of 3, the long side of the triangle will have length $\sqrt{10}$ units, as shown in [Figure 3.32](#).

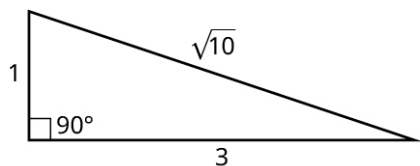


Figure 3.32 Right triangle

Of course, if we name something the real numbers, there must be numbers that aren't real. Otherwise, they'd just be called the numbers. One such not real number, one that cannot be a length, is $\sqrt{-1}$. It is part of a collection of numbers called the complex numbers, it is denoted with the letter i . As an extension, the square root of any negative number is not a real number, but instead a complex number.

To determine if a number is real, check to see if there are any negatives under a square root or any i 's. If there are any present, the number is not real.

EXAMPLE 3.96

Identifying Real Numbers

Determine if each of the following are real numbers:

1. $\frac{4\sqrt{3}}{7}$
2. 13.3381
3. $17\sqrt{-8}$

Solution

1. $\frac{4\sqrt{3}}{7}$ is a real number, as there are no negatives under the square roots, nor is there any factor of i .
2. 13.3381 is a rational number, and so it is a real number.
3. $17\sqrt{-8}$ is not a real number, as there is a negative number under the square root.

YOUR TURN 3.96

Determine if each of the following are real numbers:

1. $\frac{8}{15}$
2. $4 + 17i$
3. $-17\sqrt{46}$

Identifying Subsets of Real Numbers

The real numbers were built out of pieces, including integers, rational numbers, and irrational numbers. As such, the real numbers have named subsets, as shown in the table below.

Set Name	Set Symbol	Set Description
Natural Numbers	$\mathbb{N}$	The counting numbers
Whole Numbers		The counting numbers and 0
Integers	$\mathbb{Z}$	The natural numbers, their negatives, and 0
Rational Numbers	$\mathbb{Q}$	Fractions of integers
Irrational Numbers	$\mathbb{P}$	Numbers that cannot be written as a fraction of integers
Real Numbers	$\mathbb{R}$	The union of the rational and irrational numbers, all possible physical lengths, and their negatives

When we categorize numbers using these sets, we use the smallest set that they belong to. For instance, -7 is an integer, and a rational number, and a real number. The smallest set to which -7 belongs is integer, so we'd say it belongs to the integers.

We can also represent the relationships between the different sets of real numbers using set notation. All natural numbers are integers, but there are integers that are not natural numbers, so $\mathbb{N} \subset \mathbb{Z}$. Similarly, every integer is a rational number, but there are rational numbers that are not integers, so $\mathbb{Z} \subset \mathbb{Q}$. The same is true of the rational numbers and the real numbers, so $\mathbb{Q} \subset \mathbb{R}$.

There is no agreed-upon symbol for the irrational numbers. If we represent the irrationals as the set A , we should note that the following are true: $\mathbb{Q} \cup A = \mathbb{R}$ and $\mathbb{Q} \cap A = \emptyset$. Recall that this means the irrationals are the complement of the rational numbers in the universal set of real numbers.

EXAMPLE 3.97

Categorizing Numbers

Identify all subsets of the real numbers to which the following real numbers belong:

- 14
- -14.223
- $\sqrt{17}$

Solution

- 14 is a natural number, integer, and rational number.
- -14.223 is a rational number.
- $\sqrt{17}$ is an irrational number.

YOUR TURN 3.97

Identify all subsets of the real numbers to which the following real numbers belong:

- $14\sqrt{3}$
- -147
- $\frac{37}{150}$

EXAMPLE 3.98**Categorizing Numbers within a Venn Diagram**

Place the following numbers correctly in the Venn diagram (Figure 3.33).

$$-4\sqrt{2} \quad -10 \quad \frac{37}{150} \quad 41 \quad \frac{1}{20} \quad 4\pi$$

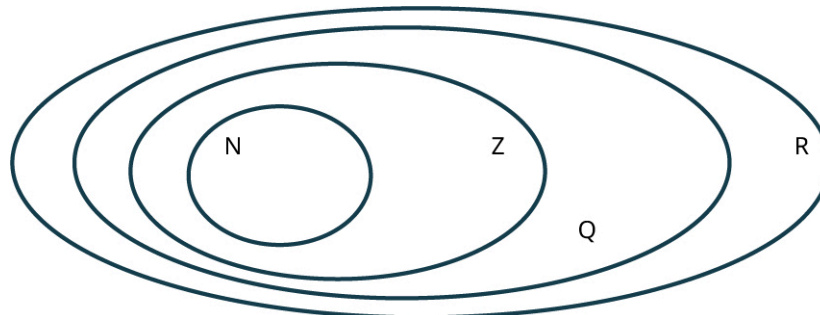


Figure 3.33

✓ Solution

Since $-4\sqrt{2}$ is irrational, it belongs in the real numbers, but outside the rational numbers (Figure 3.34).

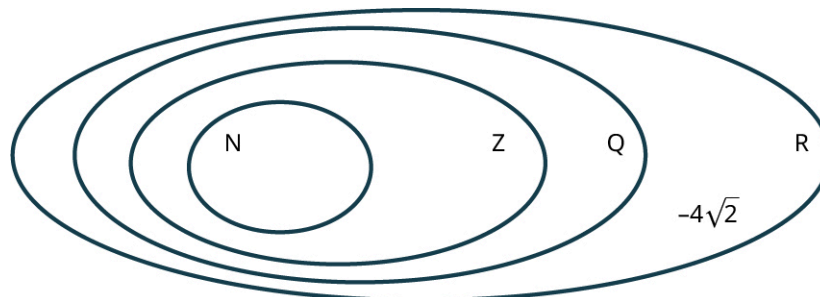


Figure 3.34

Since -10 is an integer, it belongs in the integers but outside the natural numbers (Figure 3.35).

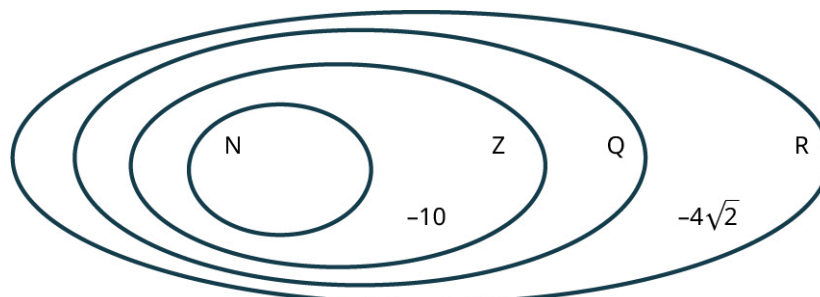


Figure 3.35

Since $\frac{37}{150}$ is a rational number, it belongs in the rational numbers but not in the integers (Figure 3.36).

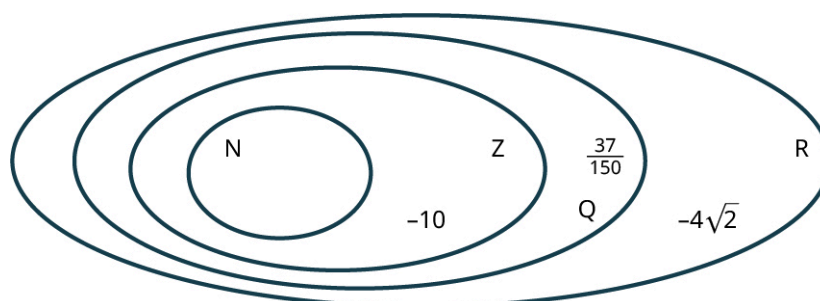


Figure 3.36

Since 41 is a natural number, it belongs in the natural numbers circle (Figure 3.37).

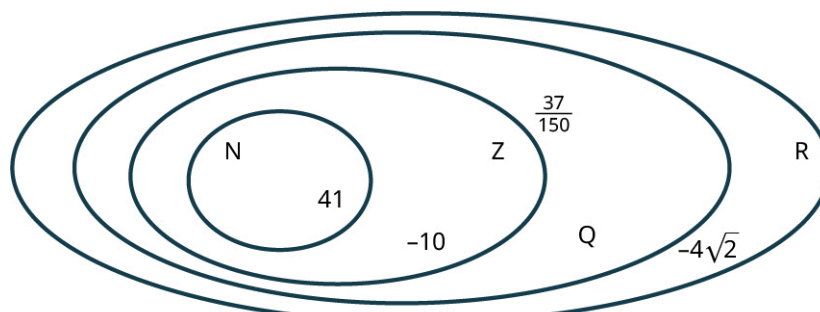


Figure 3.37

Since $\frac{1}{20}$ is a rational number, it belongs in the rational numbers but not in the integers (Figure 3.38).

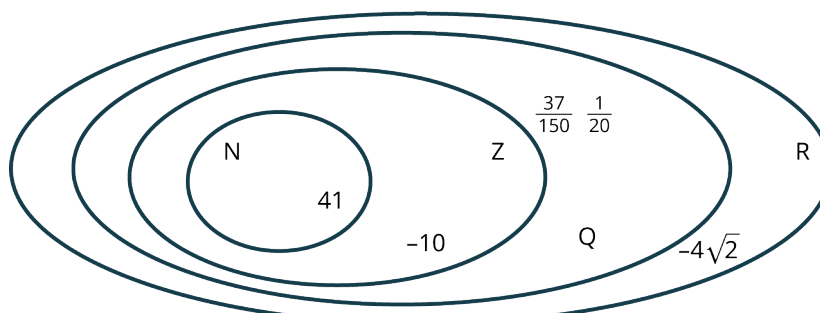


Figure 3.38

Since 4π is irrational, it belongs in the real numbers, but outside the rational numbers (Figure 3.39).

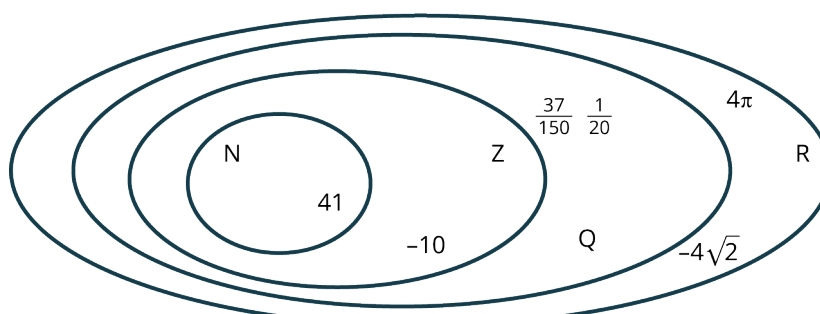
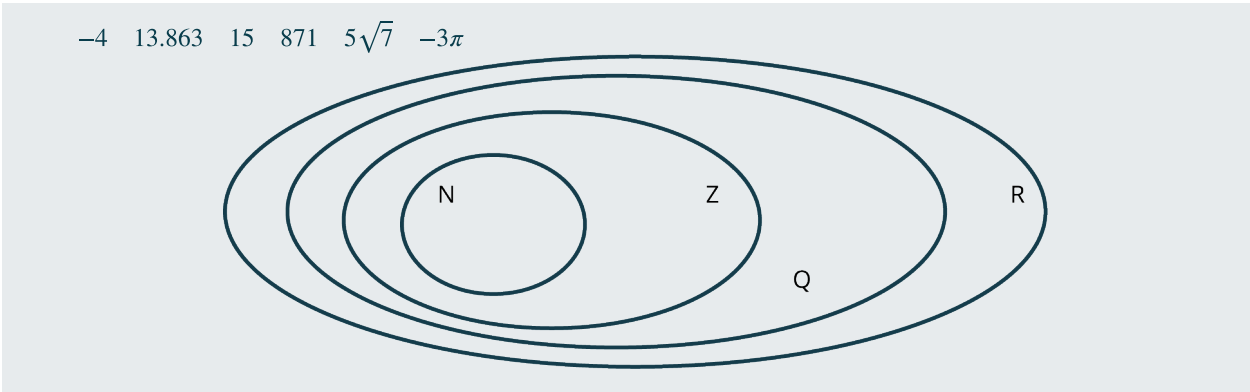


Figure 3.39

> YOUR TURN 3.98

1. Place the following numbers correctly into the Venn diagram.



▶ VIDEO

[Identifying Sets of Real Numbers \(https://openstax.org/r/Sets_of_Real_Numbers\)](https://openstax.org/r/Sets_of_Real_Numbers)

Recognizing Properties of Real Numbers

The real numbers behave in very regular ways. These behaviors are called the **properties of the real numbers**. Knowing these properties helps when evaluating formulas, working with equations, or performing algebra. Being familiar with these properties is helpful in all settings where numbers are used and manipulated. For example, when multiplying $4 \times 13 \times 25$, you could multiply the 4 and 25 first. If you know that product is 100, it makes the multiplication easier.

The table below is a partial list of properties of real numbers.

Property	Example	In Words
Distributive property $a \times (b + c) = a \times b + a \times c$	$5 \times (3 + 4) = 5 \times 3 + 5 \times 4$	Multiplication distributes across addition
Commutative property of addition $a + b = b + a$	$3 + 7 = 7 + 3$	Numbers can be added in any order
Commutative property of multiplication $a \times b = b \times a$	$10 \times 4 = 4 \times 10$	Numbers can be multiplied in any order
Associative property of addition $a + (b + c) = (a + b) + c$	$4 + (3 + 8) = (4 + 3) + 8$	Doesn't matter which pair of numbers is added first
Associative property of multiplication $a \times (b \times c) = (a \times b) \times c$	$2 \times (5 \times 7) = (2 \times 5) \times 7$	Doesn't matter which pair of numbers is multiplied first
Additive identity property $a + 0 = a$	$17 + 0 = 17$	Any number plus 0 is the number
Multiplicative identity property $a \times 1 = a$	$21 \times 1 = 21$	Any number times one is the number
Additive inverse property $a + (-a) = 0$	$14 + (-14) = 0$	Every number plus its negative is 0
Multiplicative inverse property $a \times (\frac{1}{a}) = 1$, provided $a \neq 0$	$3 \times (\frac{1}{3}) = 1$	Every non-zero number times its reciprocal is 1

The names of the properties are suggestive. The **commutative properties**, for example, suggest commuting, or moving. **Associative properties** suggest which items are associated with others, or if order matters in the computation. The **distributive property** addresses how a number is distributed across parentheses.

EXAMPLE 3.99

Identifying Properties of Real Numbers

In each of the following, identify which property of the real numbers is being applied.

1. $4 + (8 + 13) = (4 + 8) + 13$
2. $34 \times \left(\frac{1}{34}\right) = 1$
3. $14 + 27 = 27 + 14$

✓ Solution

1. Here, the pair of numbers that is added first is switched. This is the associative property of addition.
2. Here, a number is multiplied by its reciprocal, resulting in 1. This is the multiplicative inverse property.
3. Here, the order in which numbers are added is switched. This is the commutative property of addition.

> YOUR TURN 3.99

In each of the following, identify which property of the real numbers is being applied.

1. $5 \times (6 + 19) = 5 \times 6 + 5 \times 19$
2. $41.7 + (-41.7) = 0$

Using these properties to perform arithmetic quickly relies on spotting easy numbers to work with. Look for numbers that add to a multiple of 10, or multiply to a multiple of 10 or 100.

EXAMPLE 3.100

Using Properties of Real Numbers in Calculations

Use properties of the real numbers and mental math to calculate the following:

1. $2 \times 13 \times 50$
2. $13 + 84 + 27$
3. $9 \times 16 \times 11$

✓ Solution

1. Notice that $2 \times 50 = 100$, so that becomes the multiplication to do first. Use the commutative property of multiplication to change the order of the numbers being multiplied.

$$2 \times 13 \times 50 = 2 \times 50 \times 13 = 100 \times 13 = 1,300$$
2. Notice that $13 + 27 = 40$, so that becomes the addition to do first. Use the commutative property of addition to change the order in which the numbers are added.

$$13 + 84 + 27 = 13 + 27 + 84 = 40 + 84 = 124$$
3. Notice that $9 \times 11 = 99$. Using that, the problem can be changed to 99×16 . That, however, doesn't look easy at all. But $99 = (100 - 1)$. Using the distributive property, we rewrite and expand this as

$$99 \times 16 = (100 - 1) \times 16 = 100 \times 16 - 1 \times 16 = 1,600 - 16$$
The last step is subtraction, so the final answer is 1,584. So, multiplying by 99 is the same as multiplying by 100, and then subtracting the other number once.

> YOUR TURN 3.100

1. Use properties of real numbers and mental math to calculate the following:

$$9 \times 8 \times 11$$

Check Your Understanding

33. Which of the following are real numbers: $4 + i$, $\sqrt{77}$, -19 , 38.902 ?
34. Indicate which of the sets are subsets of the others: $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{N}$, $\mathbb{Z}$.
35. Which property is demonstrated here: $3 \times 6 + 3 \times 15 = 3 \times (6 + 15)$?

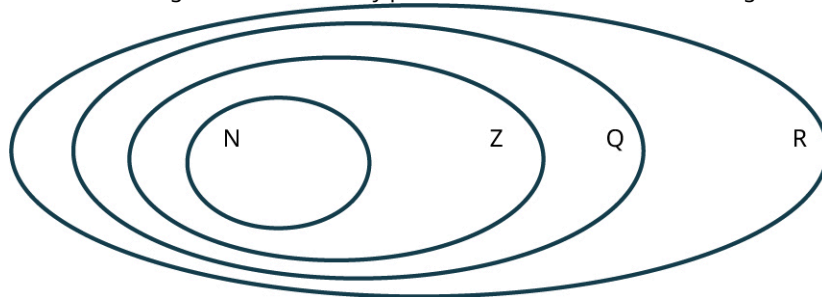


SECTION 3.6 EXERCISES

For the following exercises, identify each number as a natural number, an integer, a rational number, or a real number.

1. $\frac{1}{3}$
2. 16
3. $\sqrt{19}$
4. $-\frac{3}{7}$
5. π
6. -47
7. $\frac{13}{7}$
8. $\frac{31}{7}$

For the following exercises, correctly place the numbers in the Venn diagram.



9. -5.6 -7 $\sqrt{26}$ 41 $-\frac{13}{50}$ -46
10. -2π $-\sqrt{11}$ -17 679 $\frac{14}{37}$ $\frac{57}{151}$

For the following exercises, identify the property of real numbers that is being illustrated.

11. $14 + 38.9 = 38.9 + 14$
12. $37.12 + 98 = 98 + 37.12$
13. $38 \times 16 = 16 \times 38$
14. $6.3 \times \sqrt{14} = \sqrt{14} \times 6.3$
15. $4 + (8 + \sqrt{7}) = (4 + 8) + \sqrt{7}$
16. $13 \times (\frac{6}{11} \times 14) = (13 \times \frac{6}{11}) \times 14$
17. $(5.6 \times 8.7) \times 6 = 5.6 \times (8.7 \times 6)$
18. $(3.8 + \sqrt{5}) + 13 = 3.8 + (\sqrt{5} + 13)$
19. $4 \times (5 + \sqrt{3}) = 4 \times 5 + 4 \times \sqrt{3}$
20. $\sqrt{74} + (-\sqrt{74}) = 0$
21. $-3.4\pi \times (-\frac{1}{3.4\pi}) = 1$
22. $17 \times (\frac{1}{17}) = 1$
23. $4\sqrt{35} + 0 = 4\sqrt{35}$
24. $\sqrt{17} \times (5 + 3.6) = \sqrt{17} \times 5 + \sqrt{17} \times 3.6$
25. $(-10\pi) + 10\pi = 0$
26. $\frac{3}{\sqrt{7}} \times 1 = \frac{3}{\sqrt{7}}$
27. $3\pi + 7\sqrt{21} = 7\sqrt{21} + 3\pi$
28. $14 \times (3\sqrt{2} + 11.6) = 14 \times 3\sqrt{2} + 14 \times 11.6$